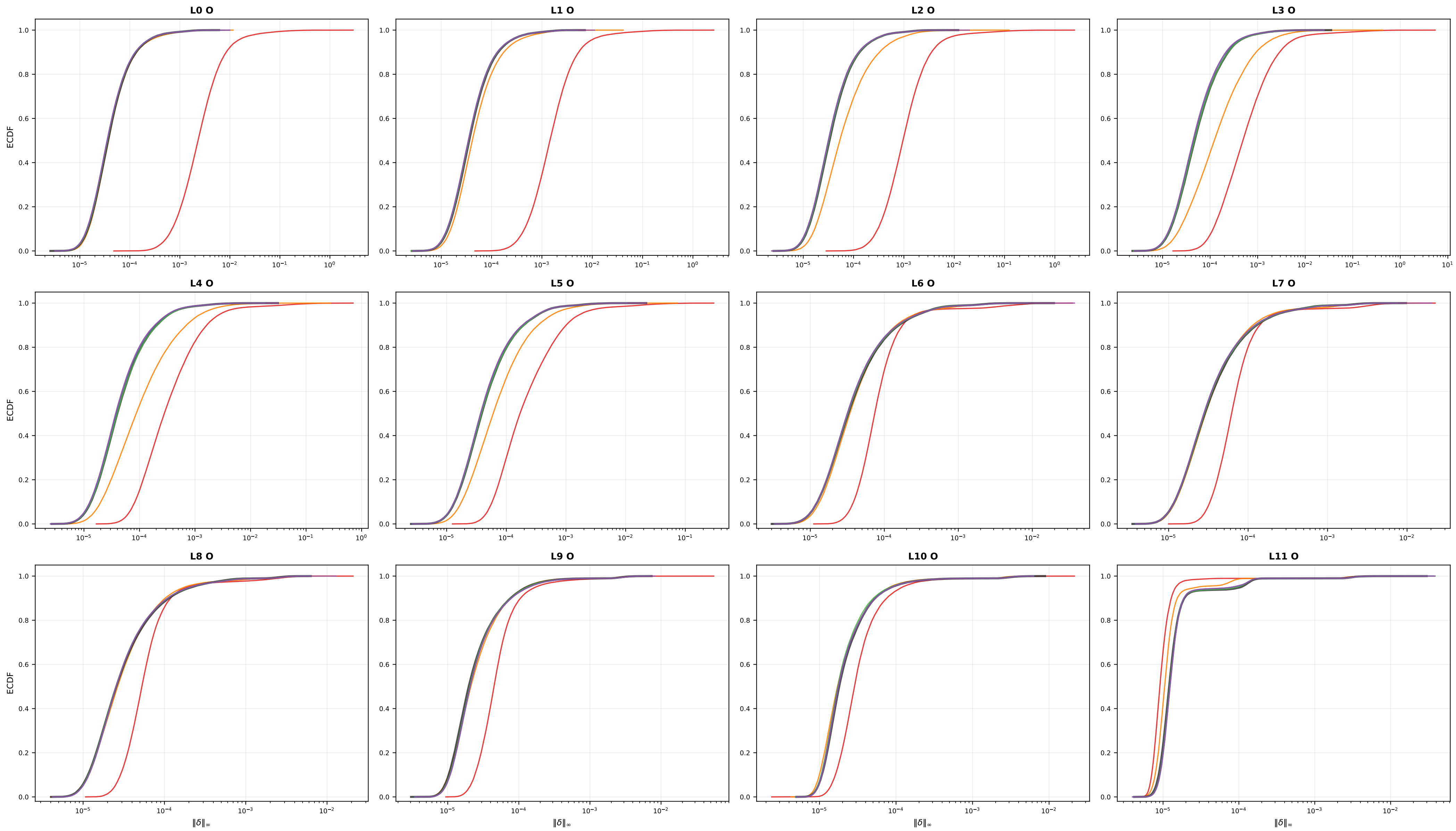


O Sublayer — Absmax ECDF per Layer × Bit Resolution
 $X = \|\delta_{vec}\|_\infty$ (raw gradient magnitude, before AbsMax NM scaling). Shape = gradient magnitude spread; different bits → different forward quantization noise



— 4-bit — 6-bit — 7-bit (baseline) — 8-bit — 10-bit — 12-bit